CUSTOMER SUPPLY CHAIN ANALYSIS REPORT

1. Executive Summary

Project Overview

The **Customer Supply Chain Dashboard** is an end-to-end Business Intelligence solution developed to analyze and monitor the operational performance of a global supply chain business. Using interactive dashboards and data visualization techniques, the project transforms raw transactional data into meaningful insights that support strategic and operational decision-making.

The solution focuses on four critical business domains: **Executive Performance, Customer Analytics, Product & Sales Analytics, and Shipping & Logistics Analysis**, enabling stakeholders to monitor key performance indicators (KPIs), identify operational inefficiencies, and uncover opportunities for business improvement.

Dataset

The project utilizes the **DataCo Global Supply Chain Dataset**, a publicly available dataset from Kaggle that simulates real-world supply chain operations across multiple countries and markets.

The dataset contains information related to:

- Customer transactions
- Product and category details
- Orders and sales
- Shipping and logistics
- Geographic locations
- Delivery performance
- Profitability metrics

The data was cleaned, normalized, and modeled into a **Star Schema** using PostgreSQL before being visualized in Power BI.

Business Value

This dashboard enables decision-makers to:

- Monitor overall sales, profit, customer growth, and operational performance.
- Identify high-performing and underperforming products.
- Analyze customer purchasing behavior and profitability.
- Evaluate shipping efficiency and delivery performance across regions.
- Detect operational bottlenecks such as delayed and cancelled shipments.
- Support data-driven decisions through interactive reports and drill-through analysis.

By consolidating business data into a single analytical platform, the dashboard improves visibility, enhances reporting efficiency, and provides actionable insights for optimizing supply chain operations.

Dashboard Solution

The Power BI solution consists of four interconnected analytical dashboards:

- **Executive Overview** – Provides a consolidated view of key business KPIs, sales trends, profit trends, market performance, and category analysis.
- **Customer Analytics** – Analyzes customer segmentation, customer profitability, geographic distribution, and purchasing trends.
- **Product & Sales Analytics** – Evaluates product performance, department revenue, category profitability, pricing, discounts, and product-level insights.
- **Shipping & Logistics Analysis** – Monitors delivery performance, shipping efficiency, delays, cancellations, shipping modes, and regional logistics performance.

The solution incorporates advanced Business Intelligence features such as **Star Schema modeling, DAX measures, Power Query transformations, drill-through pages, dynamic tooltips, cross-filtering, slicers, and interactive navigation**, delivering a scalable and user-friendly analytical experience.

2. Business Background

2.1 Business Context

Modern retail and e-commerce organizations operate complex supply chains involving customers, products, warehouses, logistics providers, and multiple regional markets. Managing these operations efficiently is critical to ensuring timely deliveries, maximizing profitability, and maintaining customer satisfaction.

This project is based on the **DataCo Global Supply Chain Dataset**, a publicly available dataset from Kaggle that simulates the day-to-day operations of a global retail supply chain. The dataset captures business activities across various geographical regions, product categories, shipping modes, and customer segments, making it suitable for Business Intelligence and supply chain analytics.

The objective of this project is to transform raw operational data into meaningful business insights through interactive dashboards, enabling stakeholders to monitor performance and make informed decisions.

2.2 Supply Chain Process

The business process represented in the dataset follows a typical retail supply chain lifecycle.

1. Customer Places Order
2. Product Availability Check
3. Order Processing
4. Warehouse Picking & Packing
5. Shipping Method Selection
6. Shipment Dispatch
7. Product Delivery
8. Delivery Status Recorded
9. Sales & Profit Analysis

2.3 Business Workflow

The dashboard is designed to support the analytical workflow followed by business stakeholders.

Step 1 – Customer Purchase

Customers from different markets place orders for products across multiple categories and departments.

Step 2 – Order Processing

Each order is processed, assigned a transaction type, and linked with customer, product, geographic, and shipping information.

Step 3 – Shipping & Delivery

Orders are shipped using different shipping modes such as Standard Class, Second Class, First Class, and Same Day. During this stage, the organization tracks:

- Scheduled shipping days
- Actual shipping days
- Delivery status
- Late delivery risk
- Cancelled shipments

Step 4 – Revenue & Profit Generation

Once orders are completed, the business records:

- Sales revenue
- Order profit
- Product quantity sold
- Discounts
- Profit margin

These metrics are used to evaluate financial performance and identify profitable products and markets.

Step 5 – Business Intelligence & Decision Making

Finally, the processed data is analyzed through Power BI dashboards to answer key business questions such as:

- Which markets generate the highest revenue?
- Which customer segments contribute most to sales?
- Which products are most profitable?
- Where are delivery delays most frequent?
- Which shipping modes perform best?
- How do discounts impact profitability?

The insights generated from these analyses help management optimize inventory, improve logistics performance, enhance customer satisfaction, and support strategic business planning.

3. Business Problem

Business Challenges

Organizations operating global retail supply chains manage thousands of customer orders across multiple products, markets, and shipping methods every day. While large volumes of operational data are generated, it often remains distributed across transactional systems, making it difficult for decision-makers to obtain a consolidated view of business performance.

Without an integrated Business Intelligence solution, management faces several operational and strategic challenges that impact profitability, customer satisfaction, and overall supply chain efficiency.

3.1 Limited Visibility into Business Performance

Business leaders require a centralized view of key performance indicators such as sales, profit, customer growth, and order volume. Without executive-level reporting, identifying trends, monitoring performance, and making timely business decisions becomes difficult.

3.2 High Late Delivery Rate

Timely delivery is one of the most important factors influencing customer satisfaction. A significant proportion of orders in the dataset experienced late deliveries, indicating potential inefficiencies in logistics operations, carrier performance, or distribution planning.

This can result in:

- Reduced customer satisfaction
- Increased operational costs
- Higher return or cancellation rates
- Negative impact on brand reputation

3.3 Limited Understanding of Customer Performance

Customers contribute differently to business revenue and profitability. Without customer-level analysis, the organization cannot easily identify:

- High-value customers
- Profitable customer segments
- Regional customer trends
- Customers generating high sales but low profitability

This limits the effectiveness of customer retention and marketing strategies.

3.4 Product Performance and Profitability Analysis

Not all products contribute equally to business success. Some products generate high revenue, while others contribute little or produce lower profit margins due to pricing, discounts, or operational costs.

Without product-level analytics, management cannot effectively:

- Identify best-selling products
- Detect underperforming products
- Evaluate category and department performance
- Optimize inventory and pricing strategies

3.5 Shipping Performance Monitoring

The organization uses multiple shipping modes across different markets. However, without detailed logistics analysis, it is difficult to determine:

- Which shipping modes perform most efficiently
- Regional delivery performance
- Average shipping duration
- Delivery delays compared to scheduled timelines

This reduces the organization's ability to improve logistics planning and service quality.

3.6 Fragmented Operational Data

The original dataset contains customer, product, sales, shipping, and geographic information within a single transactional structure. This makes reporting inefficient and increases the complexity of analytical queries.

To overcome this challenge, the data was transformed into a structured **Star Schema**, enabling faster reporting, improved scalability, and more efficient analytical processing in Power BI.

Problem Statement

The organization requires an integrated Business Intelligence solution capable of transforming raw supply chain transaction data into meaningful business insights. The solution should enable stakeholders to monitor sales performance, customer behavior, product profitability, and logistics efficiency through interactive dashboards, helping management make faster, data-driven decisions to improve overall business performance.

6. Data Preparation & ETL

6.1 Overview

The original **DataCo Global Supply Chain Dataset** consisted of a single flat table containing customer, product, order, shipping, geographic, and financial information. Although suitable for transactional storage, this structure was not optimized for analytical reporting because of repeated data, redundant attributes, and poor scalability for Business Intelligence.

To improve reporting performance and support dimensional modeling, an Extract, Transform, and Load (ETL) process was performed using **Power Query**, resulting in a normalized star schema optimized for Power BI.

6.2 Data Cleaning

Before loading the dataset into the analytical model, several data quality improvements were performed.

The following cleaning activities were completed:

- Validated column names and standardized naming conventions.
- Removed duplicate records where applicable.
- Verified customer, product, and order identifiers.
- Checked for inconsistent categorical values.
- Standardized text fields to ensure consistent reporting.
- Removed unnecessary columns that were not required for analysis.

6.3 Handling Missing Values

The dataset was examined for missing and incomplete values before modeling.

The following actions were taken:

- Reviewed null values across all business-critical columns.
- Retained records where missing values did not impact analysis.
- Replaced or transformed missing values where appropriate using Power Query.
- Ensured mandatory business fields such as Order ID, Customer ID, Product ID, Sales, Profit, and Shipping information remained valid.

6.4 Data Type Conversion

Accurate data types were assigned to improve calculation accuracy and report performance.

The following conversions were applied:

Data Type	Examples
Date	Order Date, Shipping Date
Whole Number	Customer ID, Product ID, Quantity
Decimal Number	Sales, Profit, Product Price, Discount
Text	Customer Name, Category, Department, Shipping Mode

6.5 Dimension Table Creation

Dimension tables were created to store descriptive business information used for filtering and categorization.

The following dimensions were implemented:

Dimension	Purpose
Dim Date	Calendar hierarchy and time intelligence
Dim Customer	Customer details and segmentation

Dim Product	Product information
Dim Category	Product category classification
Dim Department	Department hierarchy
Dim Geography	Market, country, region, state, and city
Dim Shipping	Shipping mode and delivery information

6.6 Fact Table Creation

A central **Fact Order Item** table was created to store measurable business transactions.

The fact table includes key business metrics such as:

- Sales
- Profit
- Quantity Sold
- Discount
- Product Price
- Order Item Total
- Foreign Keys linking all dimension tables

6.7 Power Query Transformations

After importing the normalized tables into Power BI, additional transformations were performed using Power Query.

The transformation process included:

- Renaming columns for readability.
- Assigning appropriate data types.
- Creating calculated columns where necessary.
- Removing unused fields.
- Validating relationships between tables.
- Preparing the data model for DAX calculations and visualization.

7. Data Model

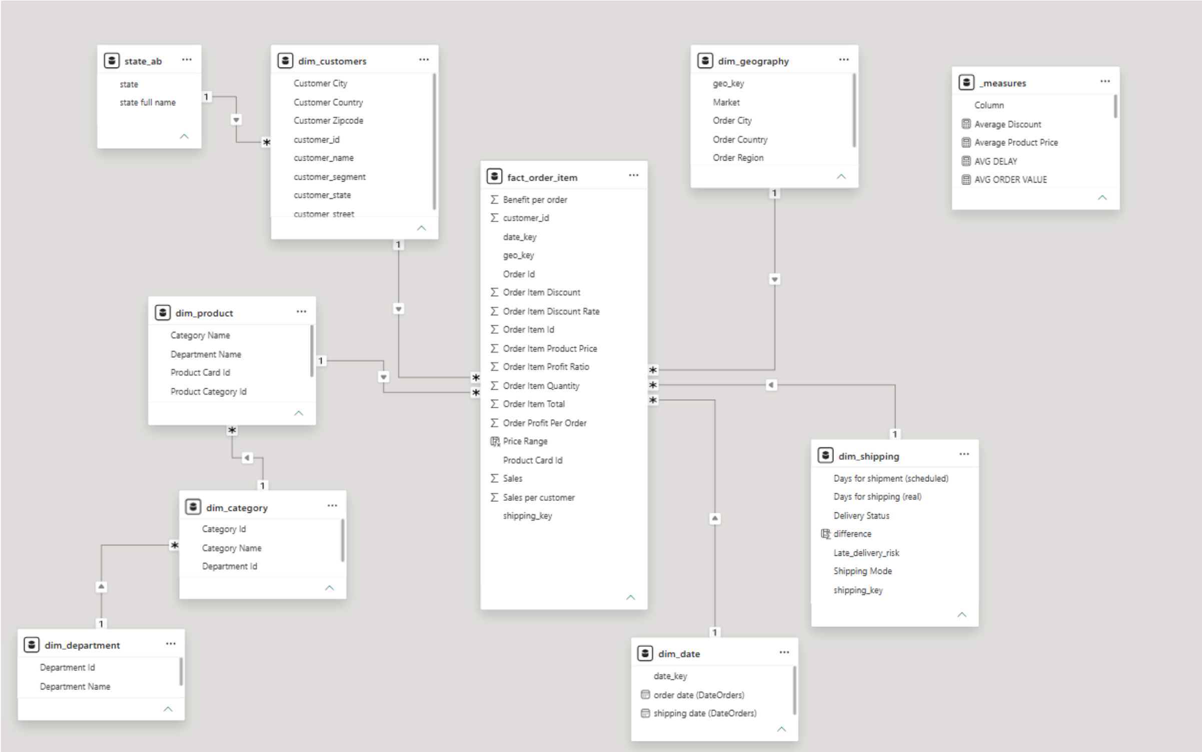


Figure 7.1: Star Schema Data Model used in the Customer Supply Chain Dashboard.

7.1 Data Model Overview

The analytical model follows a **Star Schema**, a widely adopted dimensional modeling approach in Business Intelligence. The model consists of one central fact table connected to multiple dimension tables through one-to-many relationships.

This design separates transactional data from descriptive business attributes, resulting in a scalable, maintainable, and high-performance reporting model.

7.2 Fact Table

Fact Order Item

The **Fact Order Item** table serves as the central table of the model and records every order item transaction.

7.3 Dimension Tables

The model contains multiple dimension tables that describe different aspects of each transaction.

Dim Date

Provides calendar attributes such as Year, Quarter, Month, and Date, supporting time intelligence and trend analysis.

Dim Customer

Stores customer identifiers and descriptive information used for customer segmentation and behavioral analysis.

Dim Product

Contains product-level information including product name and price.

Dim Category

Organizes products into business categories.

Dim Department

Represents higher-level product groupings for departmental performance analysis.

Dim Geography

Contains hierarchical geographic information including Market, Region, Country, State, and City.

Dim Shipping

Stores shipping-related attributes such as shipping mode, delivery status, scheduled shipping days, and actual shipping days.

Dim Transaction Type

Stores transaction classifications used to analyze different order types.

7.4 Relationships

The model uses **one-to-many (1:*) relationships**, where each dimension table is linked to the **Fact Order Item** table using surrogate or business keys.

Dimension tables act as lookup tables, while the fact table stores the transactional records. This relationship structure enables efficient filtering, aggregation, and slicing of data across different business perspectives.

7.5 Why Star Schema?

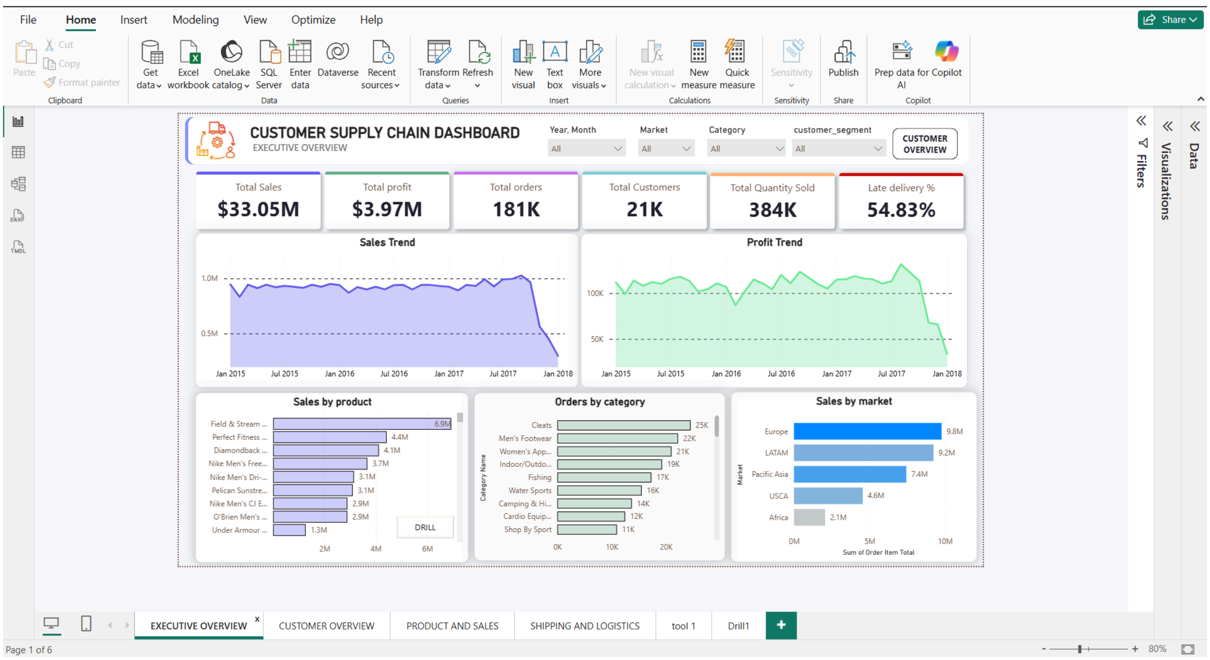
A **Star Schema** was selected because it is the industry-standard data model for analytical reporting and Business Intelligence.

Key benefits include:

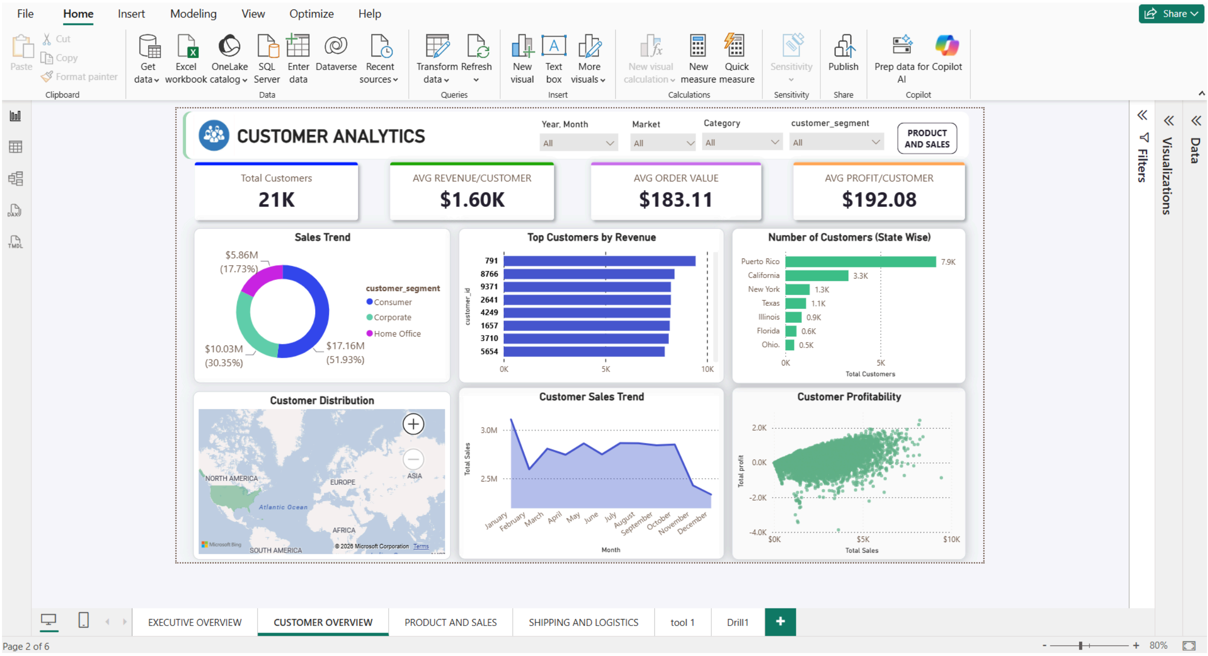
- Reduced data redundancy through normalization.
- Improved report performance by minimizing complex joins.
- Simplified DAX calculations and measure creation.
- Faster filtering and aggregation across business dimensions.
- Better scalability for future enhancements and additional data sources.
- Easier maintenance and improved readability of the data model.

8. Dashboard Overview

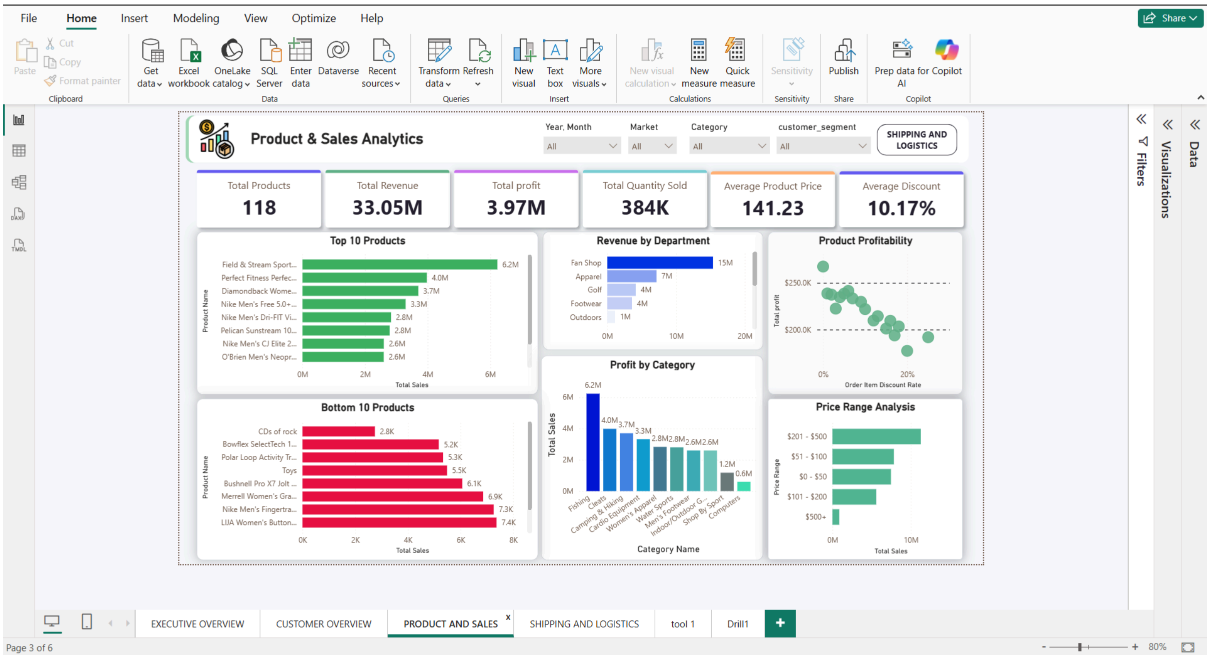
Executive Overview



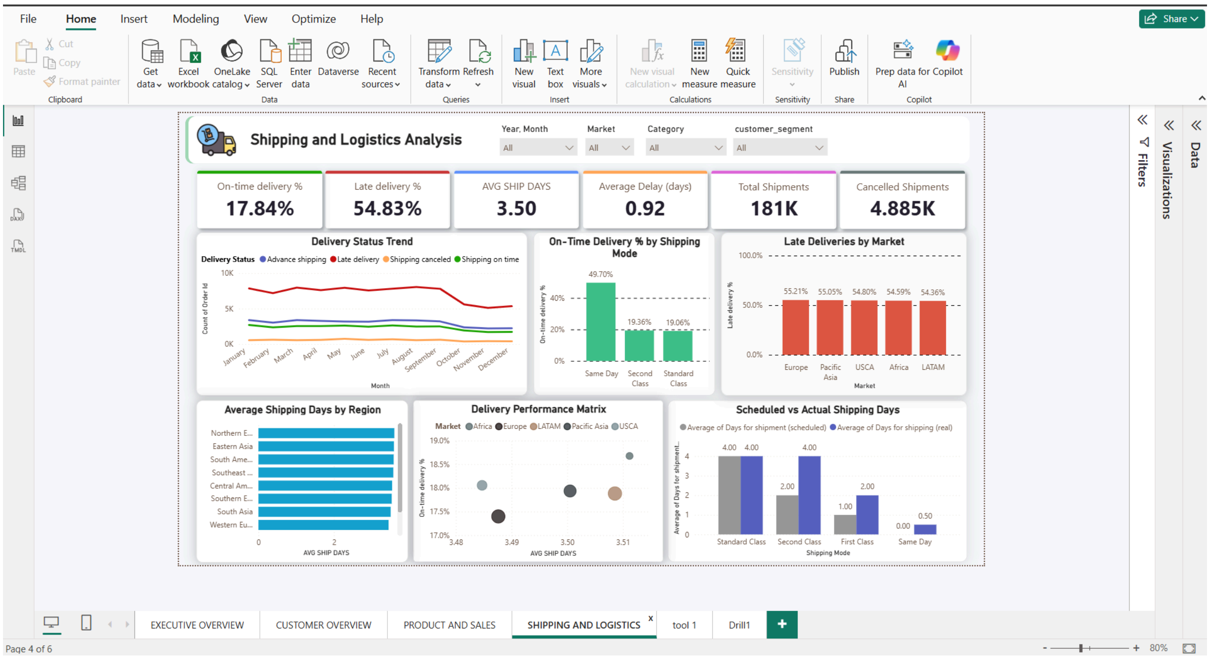
Customer Analytics



Product & Sales Analytics

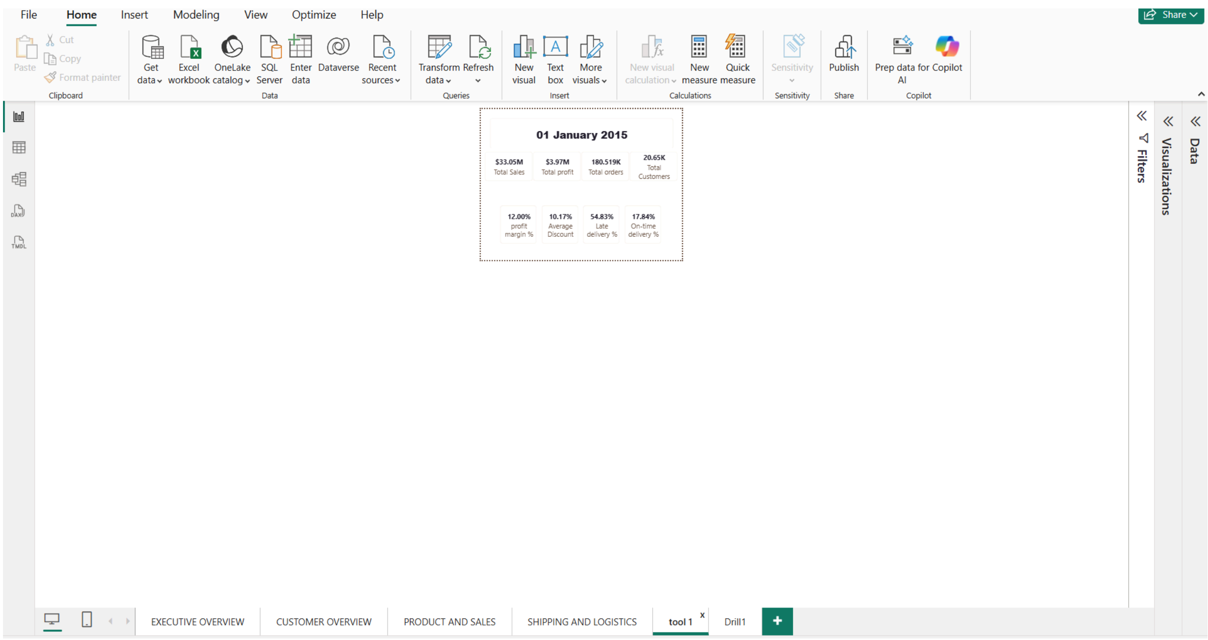


Shipping & Logistics Analysis

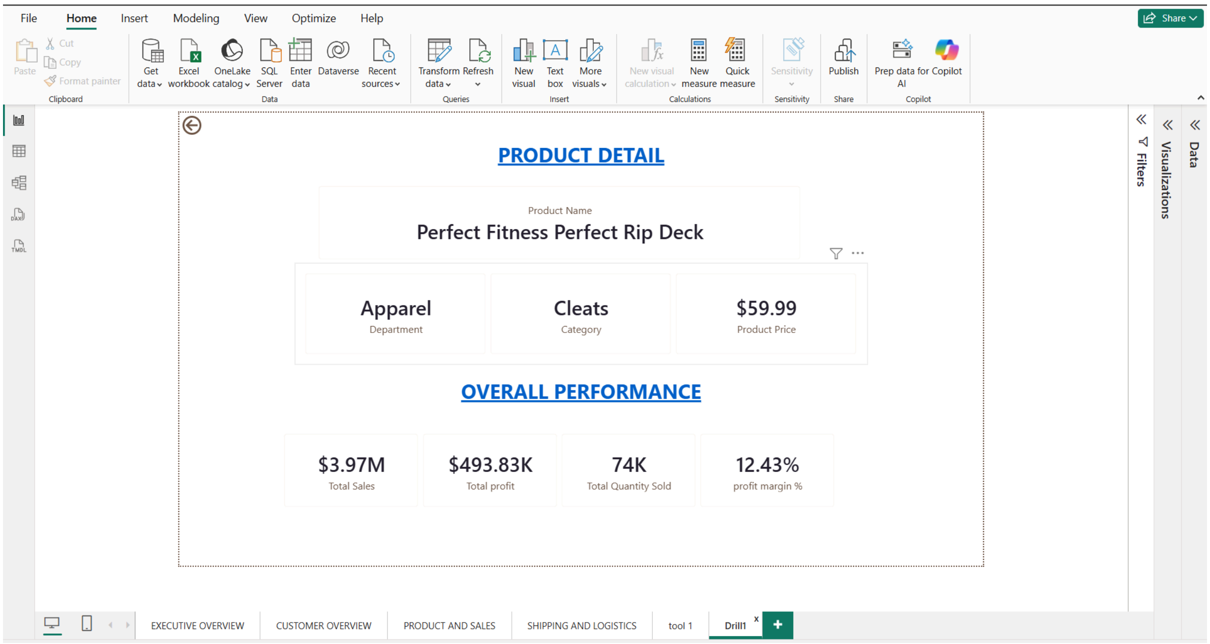


Interactive Features

Tooltip



Drill through



9. Business Insights

The dashboard provides several actionable insights into the organization's sales performance, customer behavior, product portfolio, and logistics operations. The following findings summarize the key outcomes of the analysis.

9.1 Strong Market Performance in Europe

The Europe market generated the highest sales revenue among all markets, making it the company's strongest revenue contributor. This indicates a well-established customer base and strong product demand within the region.

Business Impact:

The organization should continue investing in marketing, inventory planning, and customer retention strategies in this market while identifying opportunities to replicate its success in other regions.

9.2 Consumer Segment Drives Business Growth

The Consumer customer segment contributed the largest share of total sales, indicating that individual consumers represent the primary source of business revenue.

Business Impact:

Understanding purchasing patterns within this segment can help optimize promotional campaigns, loyalty programs, and personalized marketing initiatives.

9.3 Fishing Category Leads Product Sales

Among all product categories, Fishing generated the highest sales revenue, demonstrating strong customer demand and market acceptance.

Business Impact:

Maintaining adequate inventory levels and implementing targeted promotional campaigns for this category can help sustain revenue growth.

9.4 Product Profitability Varies Across the Portfolio

The analysis revealed that several products generated high sales volumes but comparatively lower profit margins. Higher discount rates and pricing strategies may have reduced profitability despite strong revenue generation.

Business Impact:

Product performance should be evaluated using both revenue and profit rather than sales alone to support pricing optimization and inventory decisions.

9.5 Customer Profitability Differs Significantly

Customer analysis showed that while some customers contribute substantial revenue, they generate relatively low or even negative profit. This suggests variations in discount levels, product mix, or operational costs.

Business Impact:

Customer segmentation based on profitability can help prioritize high-value customers while reviewing pricing strategies for low-margin accounts.

9.6 Delivery Performance Requires Improvement

More than half of the analyzed orders experienced late deliveries, highlighting opportunities to improve logistics planning and operational efficiency.

Business Impact:

Reducing delivery delays can improve customer satisfaction, strengthen brand reputation, and reduce costs associated with complaints, returns, and service recovery.

9.7 Shipping Mode Performance Differs

Shipping performance varied across different shipping modes. Faster shipping methods generally demonstrated better on-time delivery performance, while standard shipping methods handled larger shipment volumes.

Business Impact:

The organization can optimize shipping strategies by balancing delivery speed, operational costs, and customer expectations.

9.8 Geographic Performance is Uneven

Sales and operational performance varied across markets and regions, indicating that business growth opportunities differ geographically.

Business Impact:

Regional performance analysis can support market expansion, targeted marketing campaigns, and more effective resource allocation.

9.9 Interactive Analytics Improves Decision-Making

The implementation of drill-through pages, dynamic tooltips, cross-filtering, and interactive slicers enables users to move from high-level KPIs to detailed operational analysis without leaving the dashboard.

Business Impact:

Interactive reporting reduces analysis time, improves data accessibility, and supports faster, evidence-based decision-making across departments.

10.Recommendations

Based on the analysis, several strategic recommendations can help improve overall business performance and operational efficiency.

Improving logistics operations should be a priority, particularly in regions with high late delivery rates. Optimizing shipping routes and evaluating carrier performance can help enhance delivery reliability and customer satisfaction.

The organization should regularly review its pricing and discount strategies to ensure that increased sales do not negatively impact profitability. Products with consistently low profit margins should be evaluated for pricing adjustments or promotional optimization.

Inventory planning should focus on high-performing products and categories to maintain product availability while reducing excess stock for low-performing items. This will improve inventory utilization and support revenue growth.

Customer engagement initiatives should prioritize high-value and profitable customer segments through targeted marketing campaigns, personalized offers, and loyalty programs to strengthen long-term customer relationships.

Finally, the organization should continue leveraging Business Intelligence dashboards to monitor key performance indicators, identify emerging trends, and support timely, data-driven decision-making across sales, product, customer, and supply chain operations.

11. Conclusion

Summarize the project by highlighting:

- Successful transformation of raw supply chain data into actionable insights.
- Interactive dashboards supporting executive decision-making.
- Business intelligence solution built using Power BI, Power Query, and DAX.
- Recommendations for future enhancements, such as predictive analytics and real-time data integration.